

AP CSP | Lesson 11: Selection

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B is correct: equality satisfies $\geq$ but not $>$. A incorrectly includes equality in $>$. C reverses the outcomes. D excludes equality from $\geq$.

Q2. Answer: C

C is correct. A confuses skipping a block with stopping the program. B ignores the false condition. D describes waiting behavior that IF does not provide.

Q3. Answer: D

D is correct: $4 \geq 4$ is true, so x becomes 7 and is displayed. A skips a true branch. B reports only the increment. C applies the increment incorrectly twice or uses a value not assigned by this code.

Q4. Answer: A

A is correct: branch choice uses the condition evaluated when the IF was reached. B wrongly reevaluates the decision after the block. C invents repetition. D gives DISPLAY a control effect it does not have.

Q5. Answer: B and C

B and C are correct. Separate IF statements make separate decisions, while IF / ELSE selects a single branch. A ignores that both independent IF conditions can be true. D invents an assignment; testing does not reset values. Award one mark only for B and C together.

Q6. Answer: C

C is correct: the first block sets 5, then the second test sees 5 and adds 2. A treats the second IF as an ELSE. B ignores both updates. D adds 2 to the original score and ignores the first assignment.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) true; output 3. (b) Output 4. The IF block is skipped, but DISPLAY is outside it and executes afterward.

1 mark: States that $6 > 4$ is true.

1 mark: Gives output 3 for the original program.

1 mark: Gives output 4 for the changed initialization.

1 mark: Explains that DISPLAY is outside the IF block and executes after the skipped block.

Q8. 4 marks

Answer: minutes ← INPUT()

```
IF(minutes >= 30)
{
  DISPLAY("Goal met")
}
ELSE
{
  DISPLAY("Keep going")
}
```

Tests: 29 → Keep going; 30 → Goal met.

1 mark: Reads the numeric minutes with INPUT before using the value.

1 mark: Uses minutes >= 30 or an equivalent correct condition.

1 mark: Uses matching IF / ELSE blocks to display the two correct messages on their corresponding paths.

1 mark: Gives both boundary outputs correctly: 29 Keep going; 30 Goal met.

Q9. 4 marks

Answer: (a) After the first block: 5. Final: 7. (b) With ELSE: 5. The separate second IF tests the updated 5; ELSE belongs to the original true decision and is skipped.

1 mark: Gives 5 after the first block.

1 mark: Gives 7 after the separate second IF.

1 mark: Gives 5 for the IF / ELSE version.

1 mark: Explains the new condition evaluation versus the branch chosen by the original evaluation.

Q10. 4 marks

Answer: (a) Input 70; required Pass, actual Retry. (b) score >= 70. (c) For example 69 → Retry.

1 mark: Selects input 70.

1 mark: Gives required Pass and actual Retry for 70.

1 mark: Writes score >= 70 or an equivalent correct condition.

1 mark: Gives a valid score below 70 with expected output Retry.

Q11. 4 marks

Answer: (a) true; true. (b) Goal met. (c) met ← minutes >= 30. This would update met to false and skip the display.

1 mark: States true immediately after assigning met.

1 mark: States that met remains true immediately before IF.

1 mark: Gives the output Goal met for the unmodified program.

1 mark: Writes met ← minutes >= 30 immediately before IF, or an equivalent reevaluation.

Q12. 4 marks

Answer: amount ← INPUT()

IF(amount >= 100)

{

 discount ← 20

}

ELSE

{

 discount ← 0

}

DISPLAY(amount - discount)

Tests: 99 → 99; 100 → 80.

1 mark: Reads amount before using it and makes the correct >= 100 decision.

1 mark: Assigns discount 20 for qualifying amounts and 0 otherwise, through correct branches or an equivalent default-and-update structure.

1 mark: Displays amount minus the selected discount after the selection.

1 mark: Gives both test outputs: 99 for input 99 and 80 for input 100.